

MLP Notation

➤ MLP Notation

1 Layers

- An MLP has layers:
 1. **Input layer** (features, *e.g.*, $x_1, x_2, \dots, x_n$)
 2. **Hidden layer(s)** (neurons that transform inputs)
 3. **Output layer** (final predictions)

We often denote the number of layers as:

L = total number of layers (input + hidden + output)
 $L = \text{total number of layers (input + hidden + output)}$

2 Neurons

- A neuron in layer l , position j is written as:

$$a_j^{[l]}$$

- a = activation (output of the neuron)
- $[l]$ = which layer
- j = which neuron in that layer

Example:

- $a_1^{[1]}$ = output of the 1st neuron in the 1st hidden layer.
 - $a_2^{[3]}$ = output of the 2nd neuron in the 3rd layer.
-

3 Weights and Bias

- Each connection has a **weight** $w_{ij}^{[l]}$:
 - i = neuron in previous layer.
 - j = neuron in current layer.
- Each neuron has a **bias** term $b_j^{[l]}$.

Together, they compute the **linear combination**:

$$z_j^{[l]} = \sum_i w_{ij}^{[l]} a_i^{[l-1]} + b_j^{[l]}$$

4 Activation

- The **activation function** $g(\cdot)$ turns the linear sum into the neuron's output:

$$a_j^{[l]} = g(z_j^{[l]})$$

Examples of g : sigmoid, ReLU, tanh, softmax.

5 Compact Notation

Instead of writing every neuron separately, we use vectors/matrices:

- $\mathbf{a}^{[l]}$ = vector of activations in layer l .
- $\mathbf{W}^{[l]}$ = weight matrix for layer l .
- $\mathbf{b}^{[l]}$ = bias vector for layer l .

Then the forward pass is:

$$\mathbf{z}^{[l]} = \mathbf{W}^{[l]} \mathbf{a}^{[l-1]} + \mathbf{b}^{[l]}$$

$$\mathbf{a}^{[l]} = g(\mathbf{z}^{[l]})$$

6 Example

Suppose we have:

- 2 inputs (x_1, x_2)
- 1 hidden layer with 2 neurons
- 1 output neuron

Notation:

- Input layer: $a^{[0]} = [x_1, x_2]$
- Hidden layer: $a_1^{[1]}, a_2^{[1]}$
- Output: $a^{[2]}$

Equations:

$$z^{[1]} = W^{[1]}a^{[0]} + b^{[1]}, \quad a^{[1]} = g(z^{[1]})$$

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}, \quad a^{[2]} = g(z^{[2]})$$

In Summary:

- Inputs = $a^{[0]}$
 - Hidden layers = $a^{[1]}, a^{[2]}, \dots$
 - Outputs = $a^{[L]}$
 - Forward pass = $a^{[l]} = g(W^{[l]}a^{[l-1]} + b^{[l]})$
-